CliniScan: Lung-Abnormality Detection on Chest X-rays using AI

Dataset Description

1. Introduction

The dataset used in this project is **VinBigData Chest X-ray Abnormalities Detection** dataset is a large-scale medical imaging dataset available on **Kaggle** for object detection tasks in chest radiographs **annotated chest X-ray (CXR) images** for the purpose of detecting **thoracic abnormalities using object detection techniques**. Chest radiography is one of the most commonly used medical imaging modalities for diagnosing lung and heart diseases. This dataset enables automated detection of abnormalities using deep learning models such as **YOLO**.

2. Dataset Source

The dataset contains medical chest X-ray images with bounding box annotations marking abnormal regions. The images were originally provided in **DICOM** (Digital Imaging and Communications in Medicine) format, which is the standard format used in medical imaging systems.

3. Image Characteristics

3.1 Image Modality

- Medical Chest X-ray images
- Posterior–Anterior (PA) view radiographs
- Single-channel grayscale images

3.2 Image Format

- Original format: DICOM (.dcm)
- Converted format: PNG (.png)

3.3 Image Resolution

- Images resized to 640 × 640 pixels
- Standardized resolution ensures consistent input to the YOLO model

4. Annotation Details

Each image contains bounding box annotations representing abnormal regions.

4.1 Annotation Format (YOLO Format)

Each annotation is stored in a .txt file corresponding to the image filename.

Each line in the file contains:

- class_id → integer representing abnormality class
- x_center → normalized center x-coordinate (0–1)
- y_center → normalized center y-coordinate (0–1)
- width → normalized bounding box width
- height → normalized bounding box height

5. Abnormality Classes

The dataset contains multiple thoracic abnormality categories such as:

- Lung Opacity
- Pleural Effusion
- Cardiomegaly
- Nodule/Mass
- Consolidation
- Atelectasis
- Pneumothorax
- Other pathological findings

Each class represents a clinically significant condition visible in chest radiographs

6. Dataset Structure

The dataset is organized into the following directory structure:

```
dataset/
|
|— images/
|   ├ — train/
|   ├ — val/
|   └ — test/
|
|— labels/
|   ├ — train/
|   ├ — val/
|   └ — test/
```

Each image file has a corresponding label file with the same name.

7. Dataset Split

The dataset is divided into:

- 70% Training set
- 20% Validation set
- 10% Test set

This split ensures:

- Proper training of the model
- Hyperparameter tuning using validation data
- Final unbiased evaluation using test data

8. Purpose of the Dataset

The dataset is used to:

- Train an object detection model (YOLO)
- Detect abnormal regions in chest X-rays
- Assist in automated medical image analysis
- Improve early diagnosis support systems